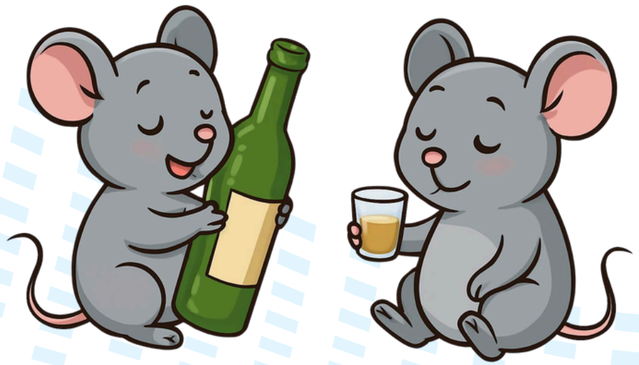
AUD in Mice Brain Regions

Analyzing 5 brain regions in mice exposed to ethanol





Deepika Chitirala, Kaye Chen, Ananya Chintalapati

Introduction of Study 01

Project Aims 02

Project Workflow 03

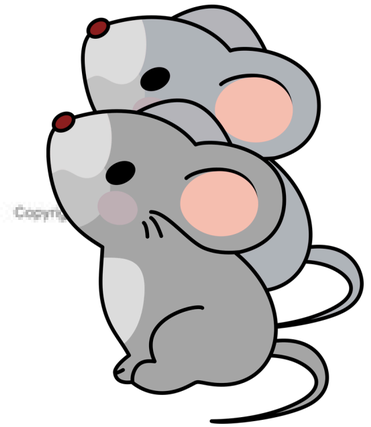
DEGs 04

KNN Classifier 05

Pathway Analysis 06

Conclusions/Results 07





Introduction



- AUD = a chronic medical condition characterized by an inability to stop or control alcohol consumption despite negative social, occupational, or health consequences

- Alcohol use disorder (AUD) has been consistently linked to transcriptomic dysregulation in the brain.

- However, little is known about how ethanol-induced transcriptomic outcomes differ between acute and chronic exposure.



Introduction cont.

Cortex

- Conscious thought
- voluntary movement
- language
- reasoning
- perception
- memory

Dorsal Hippocampus

- cognitive functions
  - spatial navigation
  - episodic memory
  - contextual learning

Ventral Hippocampus

- regulating emotion
- stress
- motivation

Dorsal Striatum

- integrates sensorimotor, cognitive, and motivational information to execute and automate behaviors
- acts as a bridge between planning and action

Ventral Striatum

- anticipating and receiving rewards
- goal-directed behavior
- learning
- reinforcement

Project Goals (Aims)



1

- **Determine the DEG for acute vs chronic AUD in each of the 5 brain regions.**
  - calculate the log2fold change and the p values

2

- **Create a KNN classifier for acute vs chronic alcohol use conditions**
  - split the data (80/20) into train and test data, and use the train data to make a classifier and test it using test data

3

- **Look at similarities and differences for which genes are expressed in specific brain regions and determine if any special genes are more DE in one region than another**
  - log 2 fold change and pathway analysis



Data

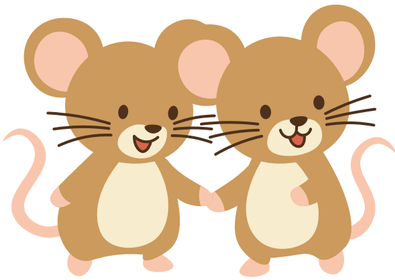
Acute Saline	Acute Ethanol	Chronic Saline	Chronic Ethanol
15	15	15	15

Cortex	Dorsal Hippocampus	Ventral Hippocampus	Dorsal Stiatum	Ventral Striatum
12	12	12	12	12

- adult, male mice, 8-10 weeks old
- 3-4 biological replicates per group
- given 2g/kg of ethanol or saline for 1 day (acute) or 10 days (chronic)

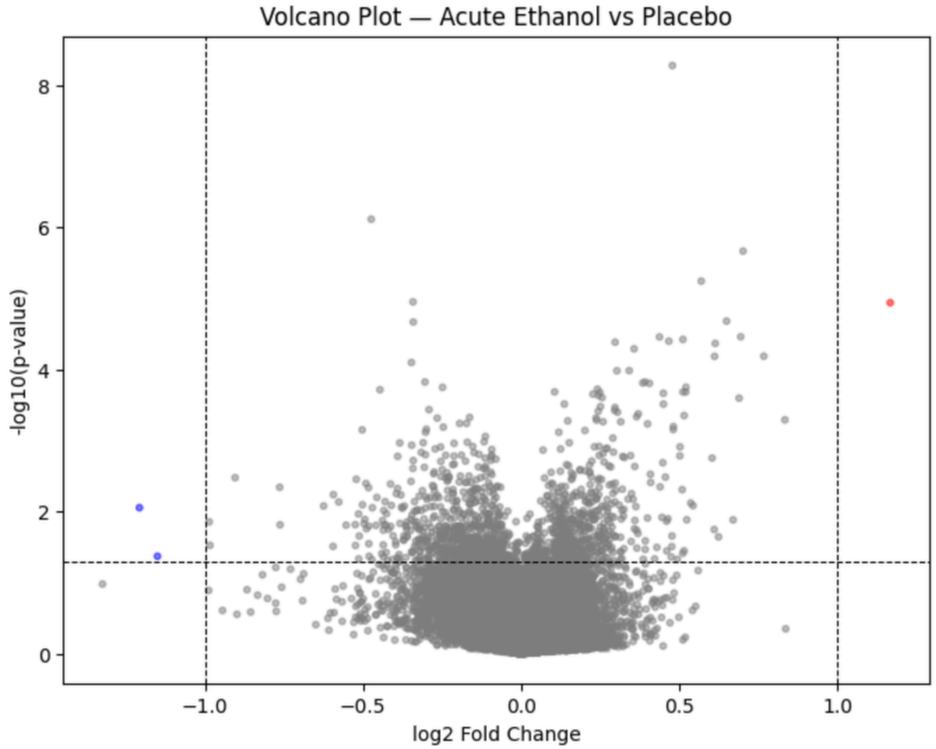


Acute and Chronic

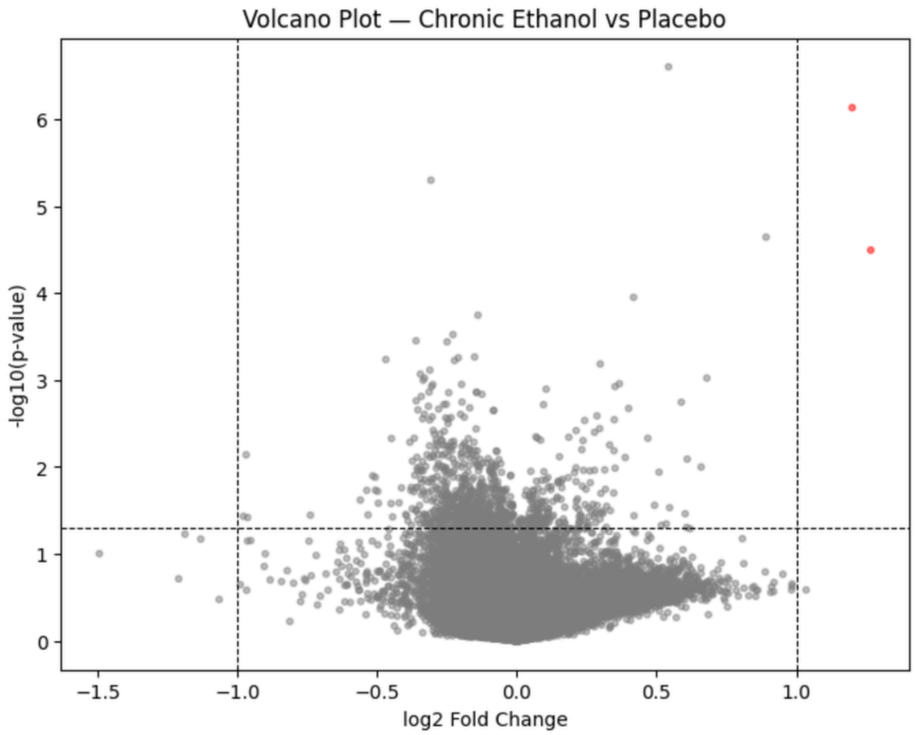


A Differential Gene Expression (DEG) analysis identifies which genes show statistically significant changes in activity (expression) between two different conditions.

x-axis: Log2 Fold Change (shows change in gene expression)
y-axis: -log10 p-value (shows statistical confidence)



Found 3 DEGs.

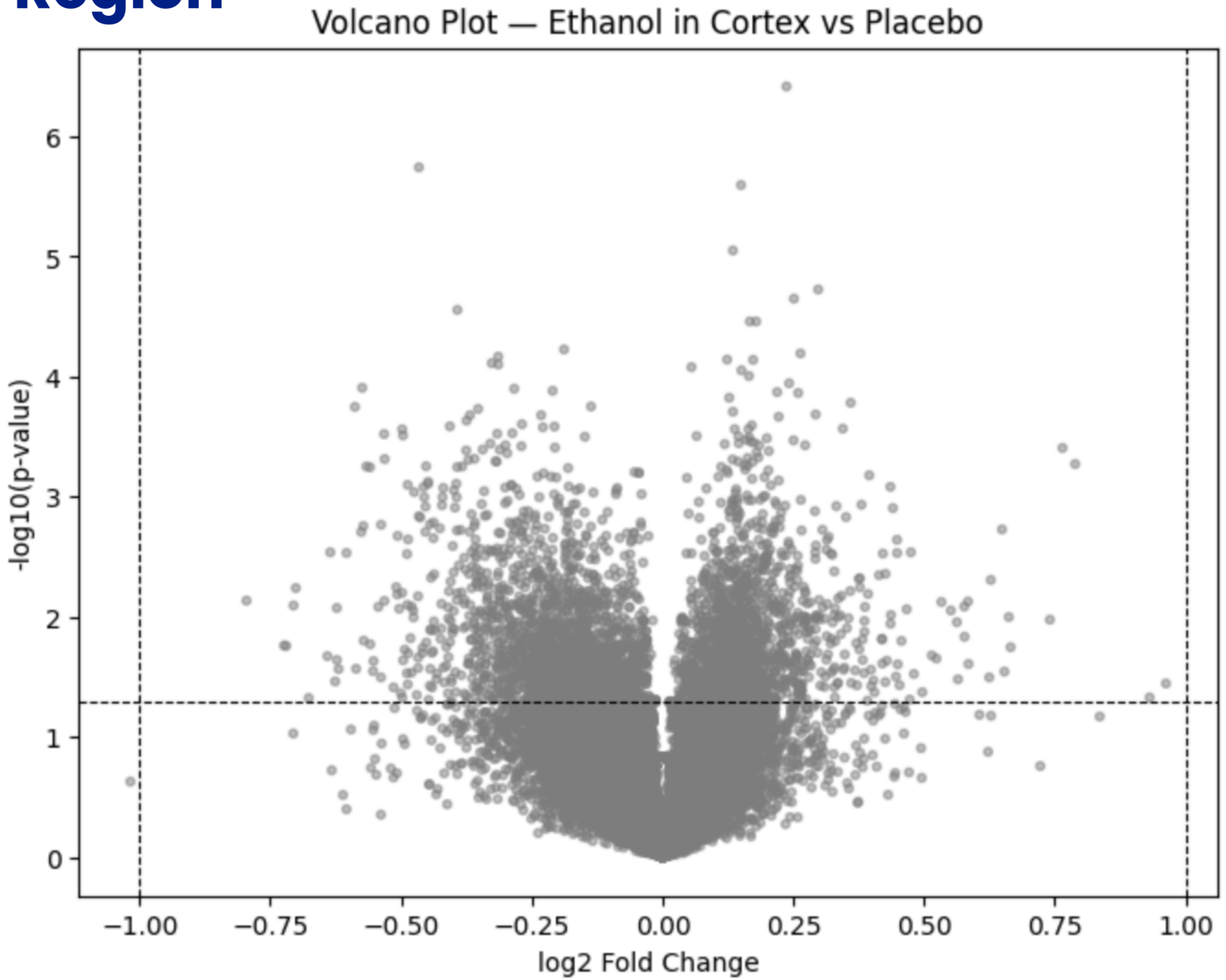


Found 2 DEGs.

Cortex - Brain Region

Found 0 DEGs





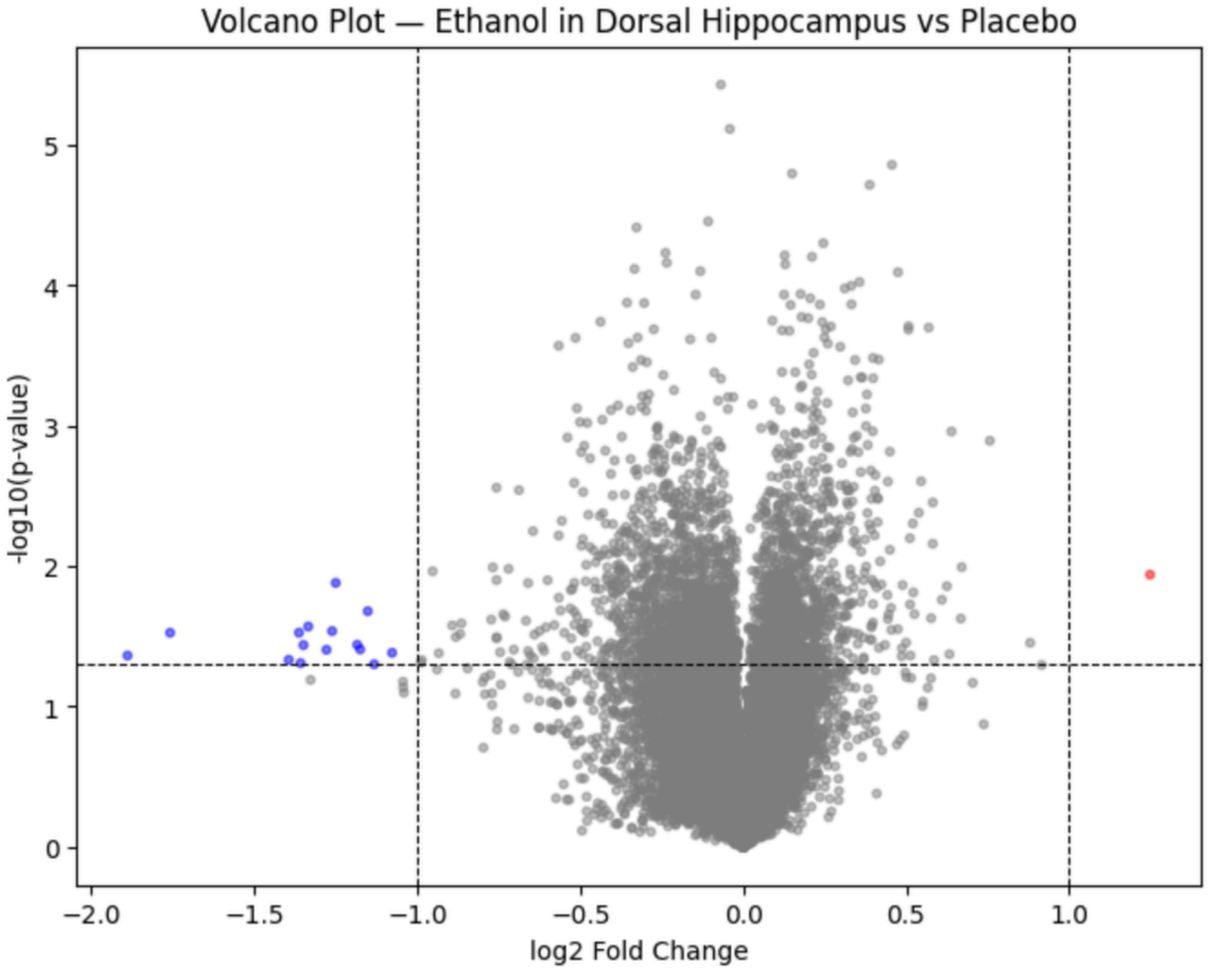
Dorsal Hippocampus

The Hippocampus is critical for learning and memory.

Found 16 DEGs.

- MoreDownregulated

Since most of the DEGs are downregulated, we can hypothesize that ethanol exposure is "turning off" or suppressing specific genetic pathways in this region, which might contribute to the memory impairments associated with alcohol usage.

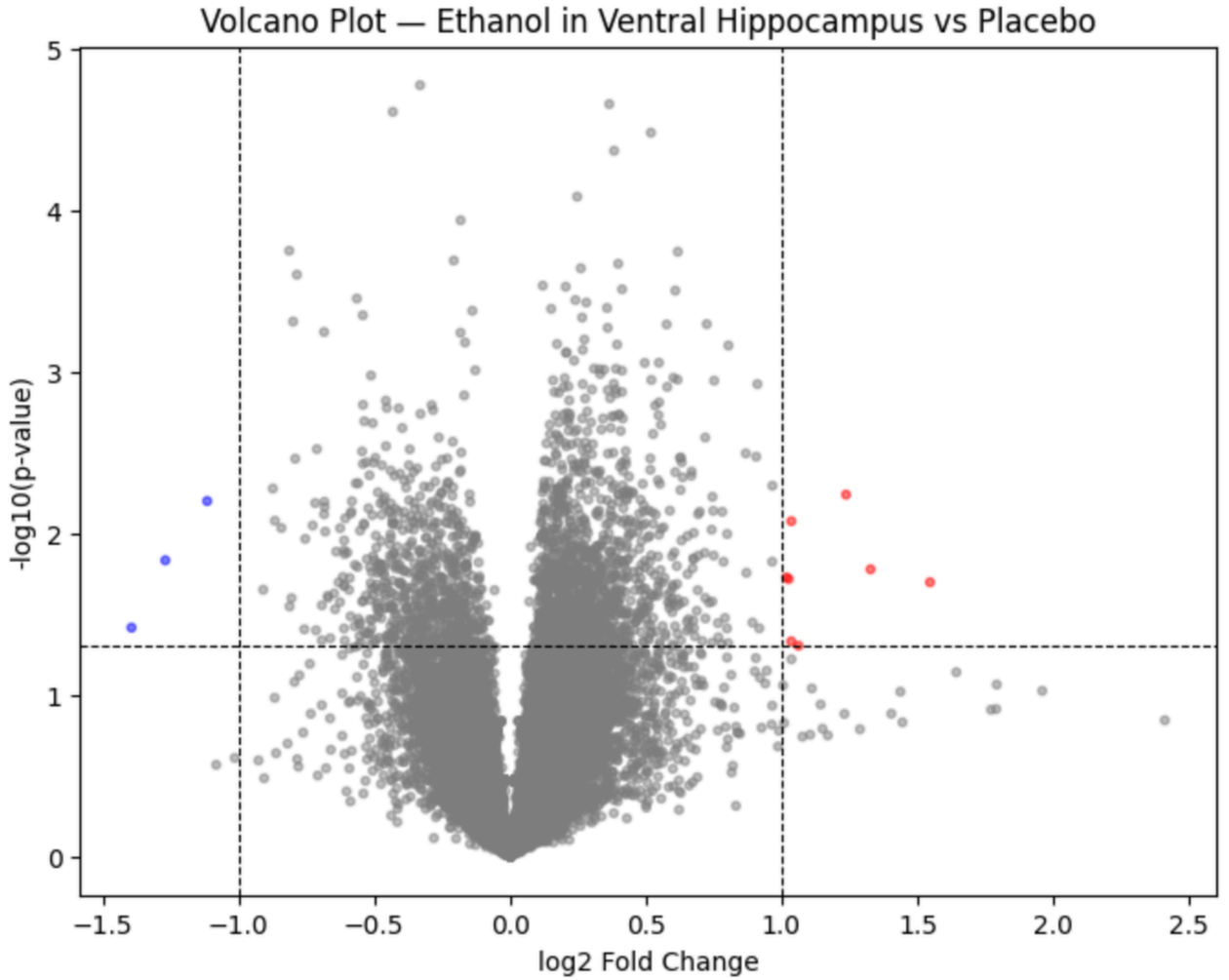


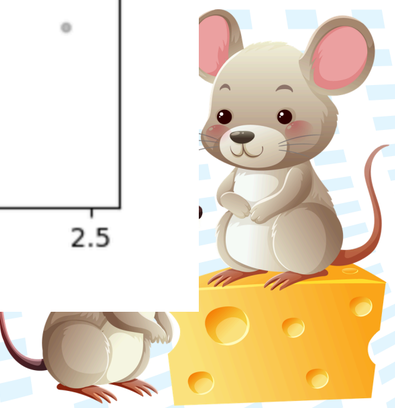


Ventral Hippocampus

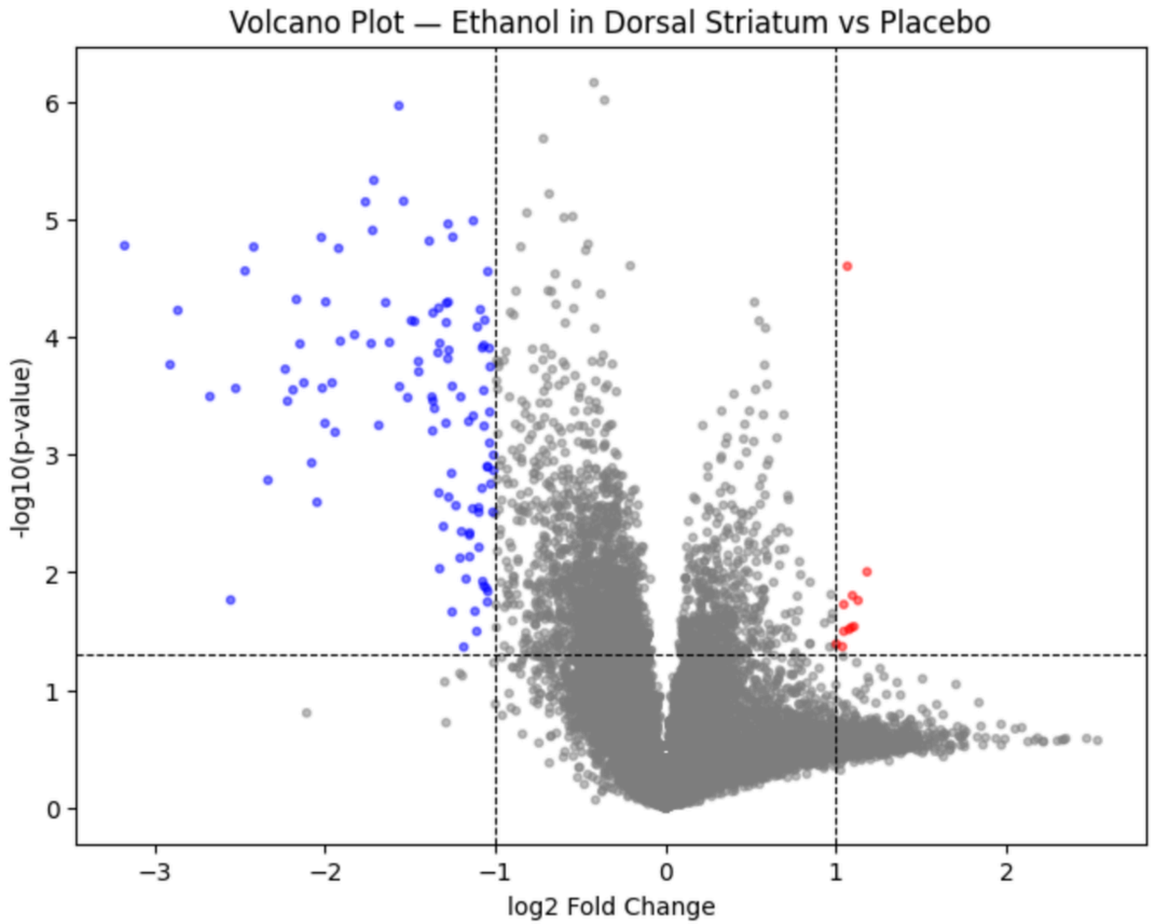
Found 11 DEGs

- More Upregulated





Dorsal Straitum



Found 118 DEGs
More Downregulated

- The Dorsal Striatum is involved in habit formation and motor control.
- Finding such a massive amount of downregulated genes here could explain the long-term changes in behavior and habit-seeking associated with AUD (Alcohol Use Disorder).



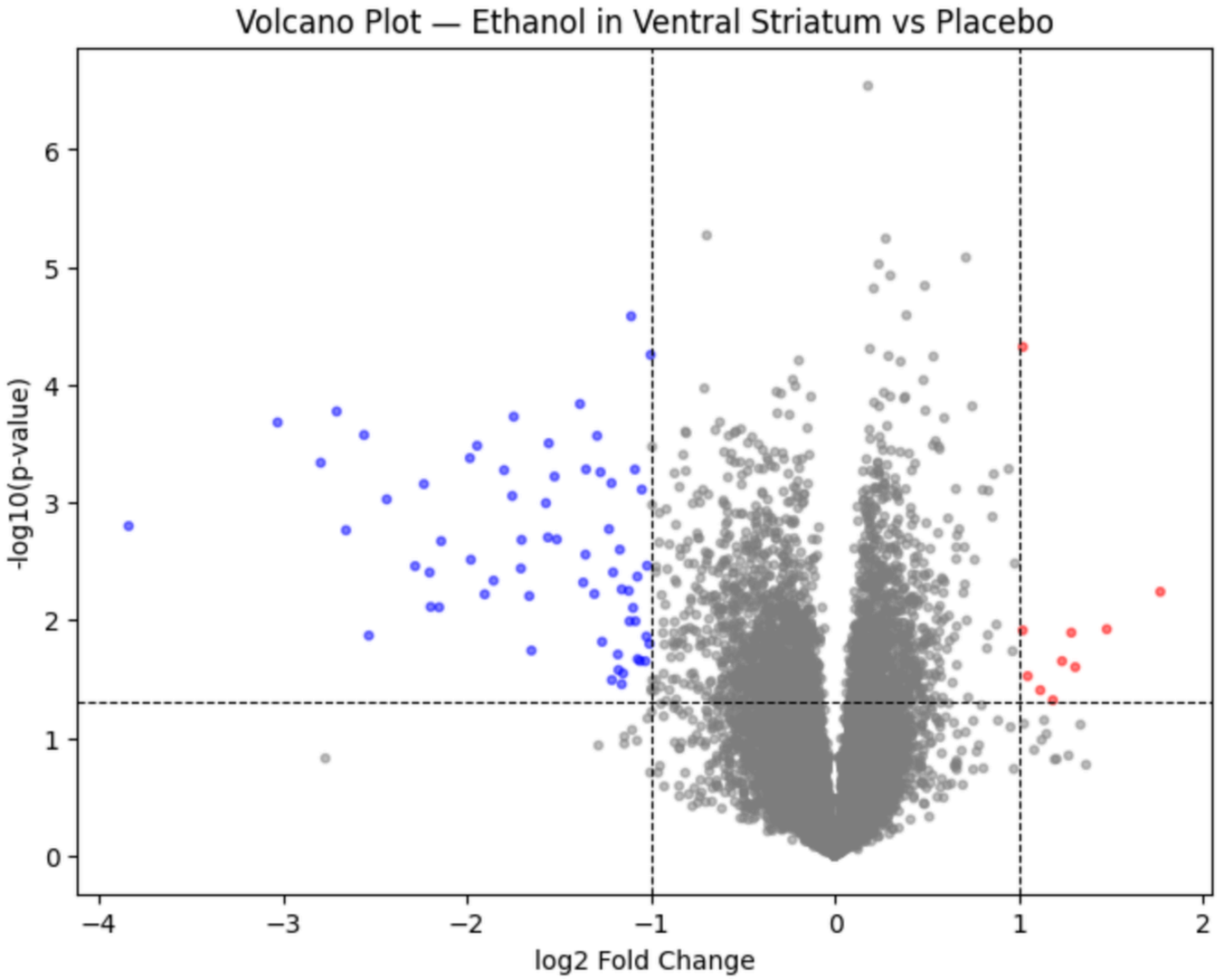
Ventral Sriatum

Found 74 DEGs

- More Downregulated

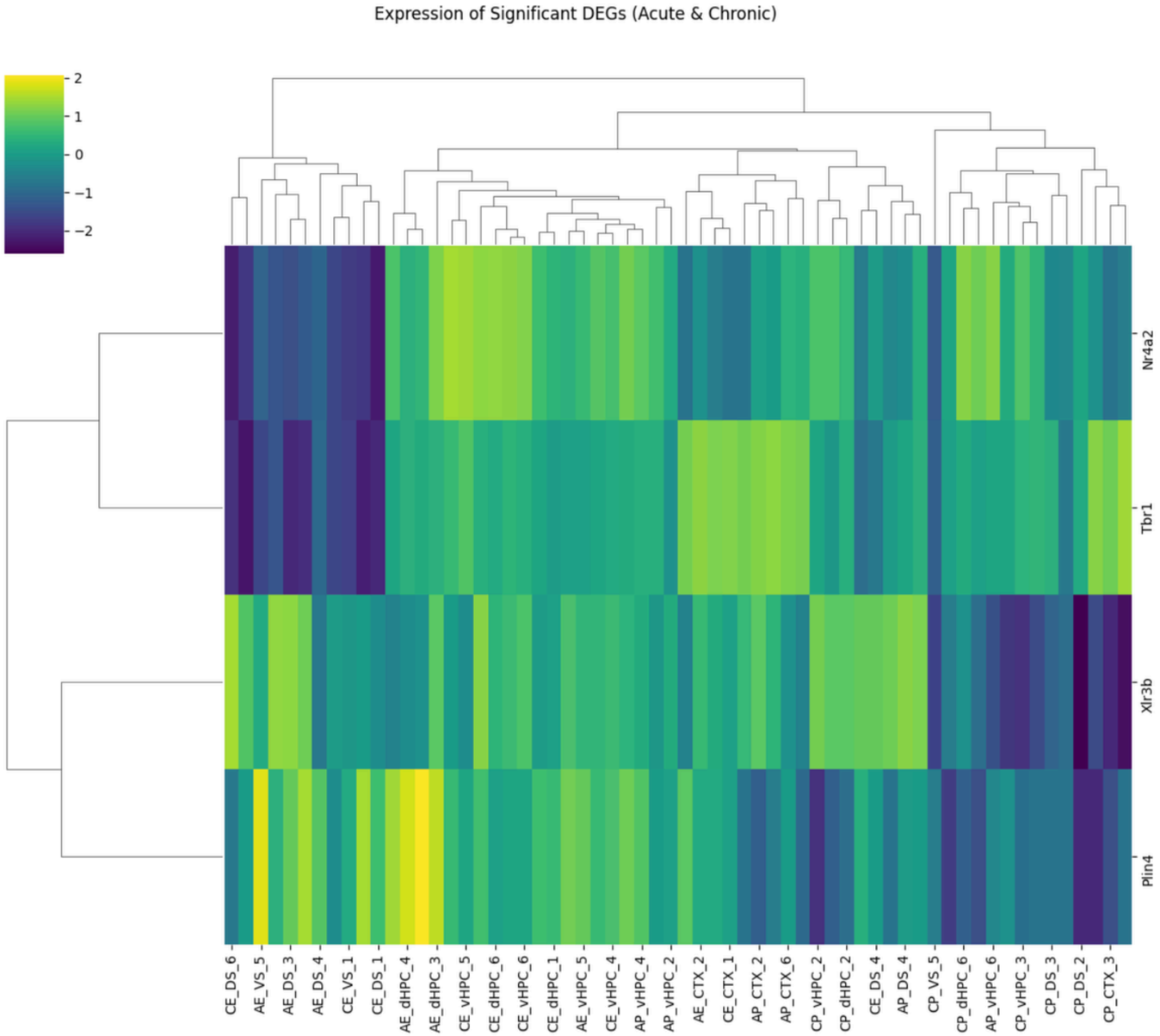
- The Ventral Striatum is a key part of the brain's reward system.
- Finding 74 significant genes here while finding zero in the Cortex, shows that ethanol exposure selectively targets specific regions involved in addiction and behavior rather than affecting the whole brain equally.





Heat Map

- **Regional Clustering:** Samples naturally group by brain region (e.g., all DS or VS samples cluster together), proving that ethanol's impact is highly region-specific.
- **Reliability:** The consistent horizontal bands of color across 3–4 biological replicates confirm that these gene changes are a stable response to ethanol rather than random noise.
- **Exposure Signatures:** The map clearly separates Acute from Chronic samples, illustrating that the duration of ethanol exposure fundamentally changes the brain's transcriptomic "fingerprint".
- **Gene Markers:** Specific genes like Nr4a2 and Plin4 show distinct "on/off" patterns across different conditions, providing clear targets for further study.

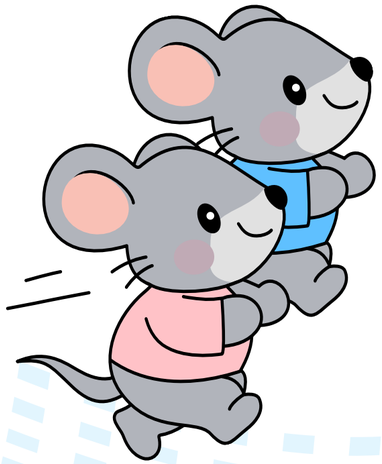


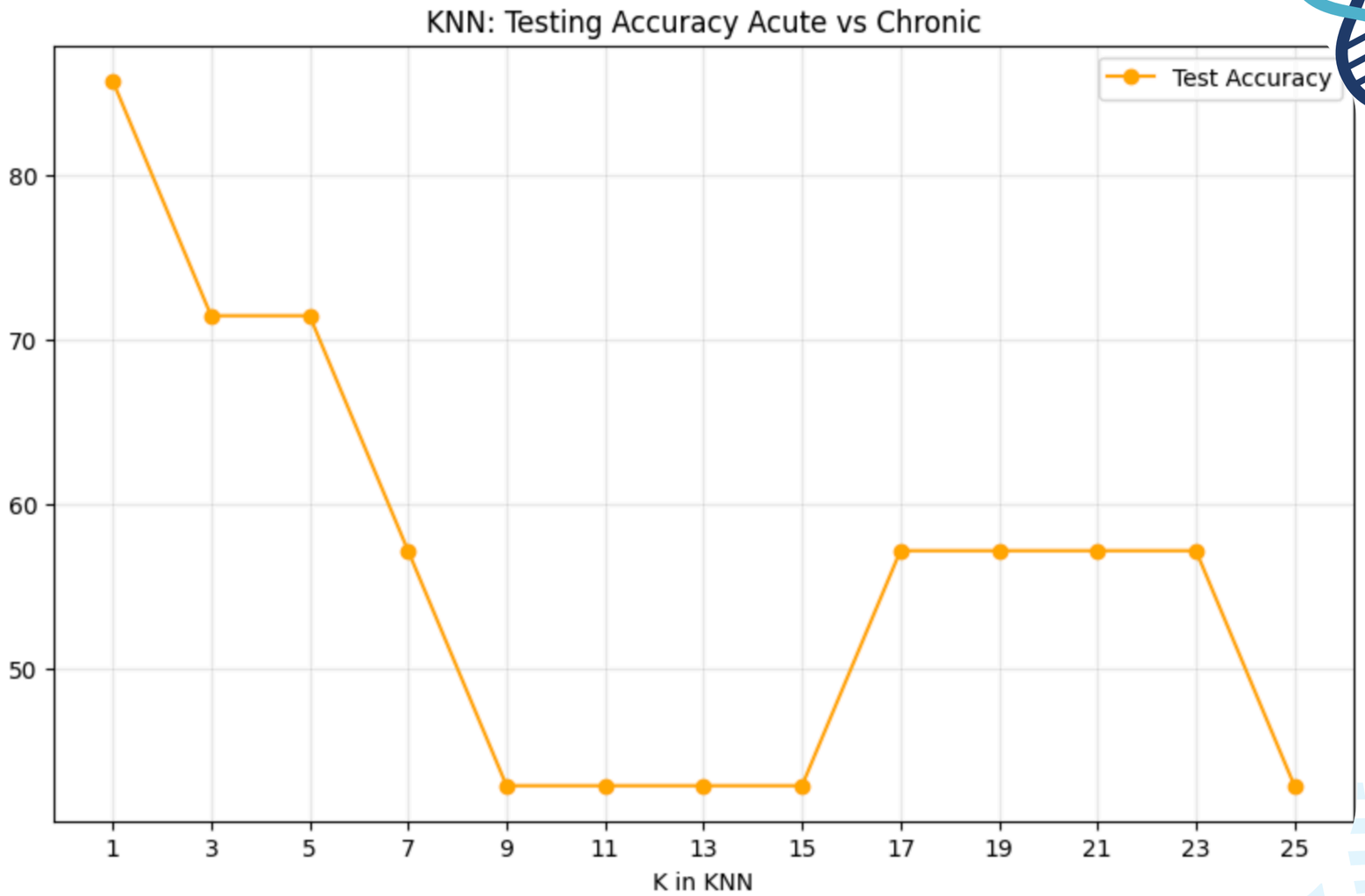
KNN Classifier

85.71%

highest

accuracy





Pathway Analysis

Dopaminergic Neurogenesis WP1498

Nuclear Receptors WP509

Dopaminergic Neurogenesis Pathway: the process where the brain builds new dopamine neurons, which are essential for movement and mood.

Acute Pathway

Gene	Expression	Role in pathway
Nr4a2	Downregulated	Tells dopamine cells to stay alive and grow
Tbr1	Downregulated	Maps where brain cells should go
Plin4	Upregulated	Coats fat droplets to protect the cell from alcohol stress

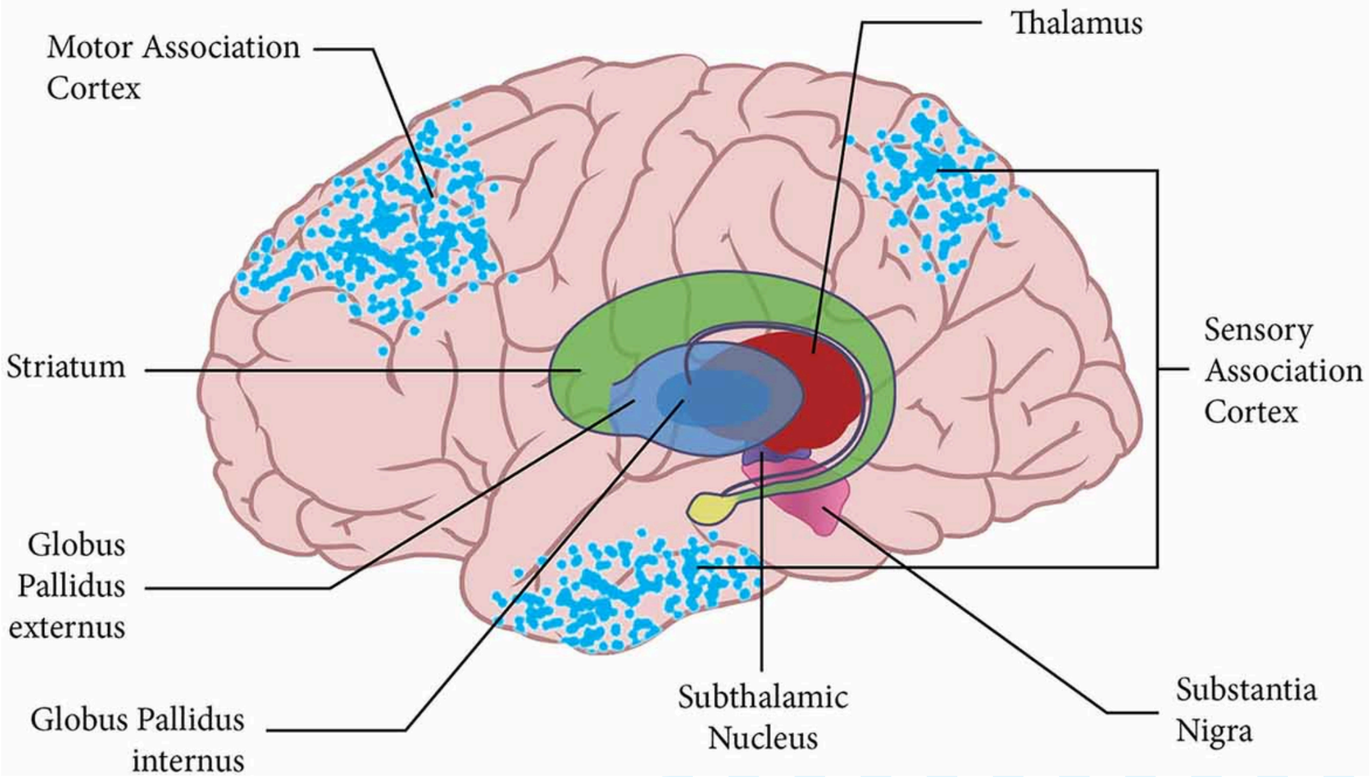
Chronic Pathway

Gene	Expression	Role in Pathway
Xlr3b	Upregulated	Helps manage thinking and memory
Plin4	Upregulated	Coats fat droplets to protect the cell from alcohol stress



Conclusion/Implications

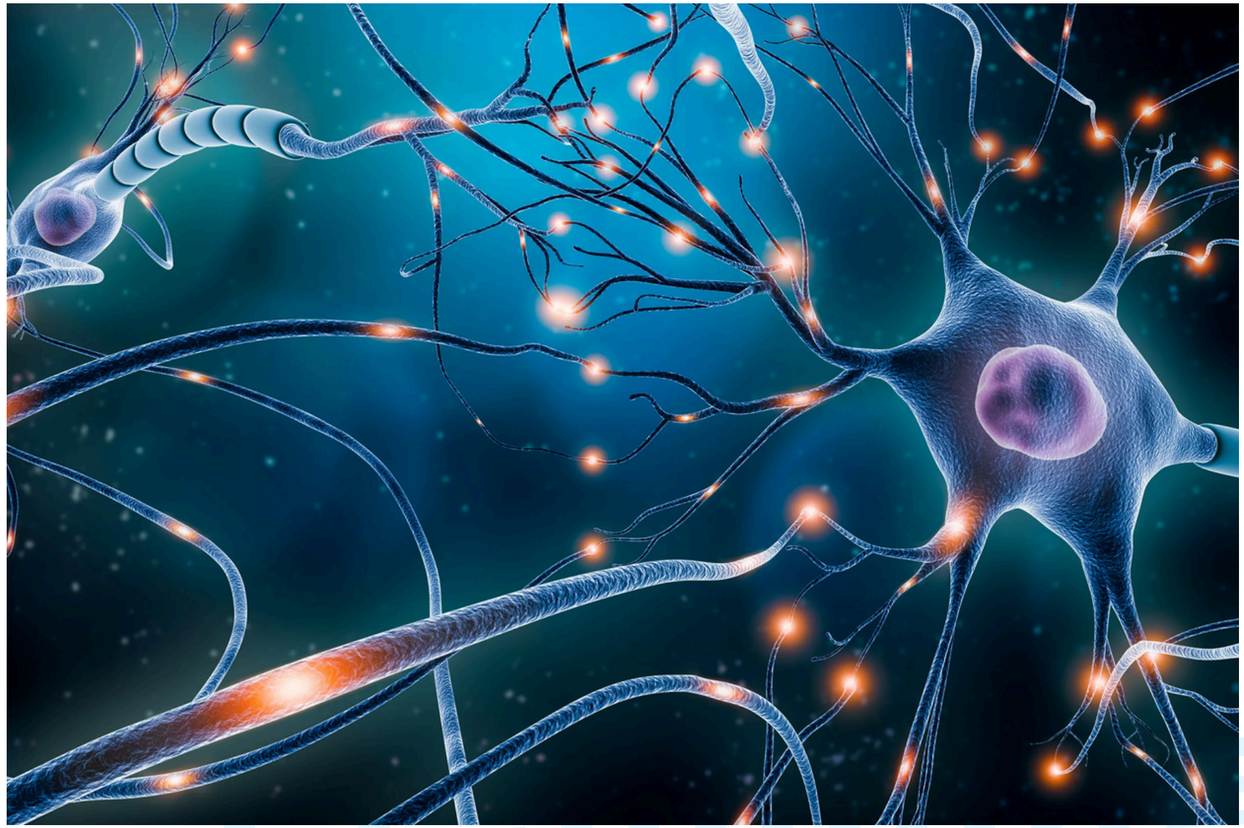
- **Acute vs Chronic**
  - Acute condition - 3 DEGs
  - Chronic condition - 2 DEGs
- **5 Brain Regions**
  - The Dorsal Striatum (118 DEGs) and Ventral Striatum (74 DEGs) showed the highest activity
  - Dorsal Hippocampus (16 DEGs) and Ventral Hippocampus (11 DEGs) showed moderate changes
  - Cortex least affected (0 DEGs)
  - Because the Striatum (the brain's reward and habit center) showed the highest activity, it suggests that alcohol rewires the brain’s motivation, indicating a shift from a choice to an automatic habit.





Conclusion/Implications

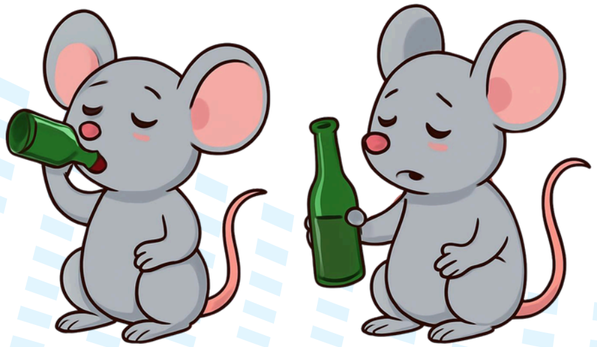
- **KNN Classifier**
  - 85.71% highest accuracy rate at K=1.
  - confirms that gene expression patterns are effective for identifying alcohol-affected states
  - could potentially be used for diagnostic tools in the future.
- **Pathway Analysis**
  - Dopaminergic Neurogenesis Pathway: the process of building new neurons.
  - The brain loses its circuit for experiencing natural pleasure.
  - implies that long-term recovery requires treatments aimed at helping the brain physically rebuild its reward circuits rather than just managing withdrawal symptoms





Thank You





References

Bamford, Ian J., and Nigel S. Bamford. “The Striatum’s Role in Executing Rational and Irrational Economic Behaviors.” The Neuroscientist, vol. 25, no. 5, 24 Jan. 2019, pp. 475–490, pmc.ncbi.nlm.nih.gov/articles/PMC6656632/#:~:text=The%20striatum%20is%20a%20cluster,%2C%20habit%20formation%2C%20and%20reward., https://doi.org/10.1177/1073858418824256.

Biotechnology, Santa. Xlr3b SiRNA (M): Sc-155381.

Cleveland Clinic. “Brain: Anatomy, Development and Function.” Cleveland Clinic, 2022, my.clevelandclinic.org/health/body/22638-brain.

Johns Hopkins Medicine. “Brain Anatomy and How the Brain Works.” Johns Hopkins Medicine, Johns Hopkins Medicine, 2025, www.hopkinsmedicine.org/health/conditions-and-diseases/anatomy-of-the-brain.

Mayo Clinic. “Alcohol Use Disorder.” Mayo Clinic, 18 May 2022, www.mayoclinic.org/diseases-conditions/alcohol-use-disorder/symptoms-causes/syc-20369243.

National Institute on Alcohol Abuse and Alcoholism. “Neuroscience: The Brain in Addiction and Recovery.” Www.niaaa.nih.gov, 22 Sept. 2023, www.niaaa.nih.gov/health-professionals-communities/core-resource-on-alcohol/neuroscience-brain-addiction-and-recovery.

Shan, Xiaoyang, et al. “Enhanced de Novo Neurogenesis and Dopaminergic Neurogenesis in the Substantia Nigra of 1-Methyl-4-Phyenyl-1,2,3,6-Tetrahydropyridine-Induced Parkinson’s Disease-like Mice.” STEM CELLS, vol. 24, no. 5, May 2006, pp. 1280–1287, https://doi.org/10.1634/stemcells.2005-0487. Accessed 18 Feb. 2022.